The Effects of Weather Conditions On Energy Generation and Consumption

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***Abstract*—Global energy policies has shifted to renewable energy resources due to the environmental effects of fossil fuels and the increasing distrust to the nuclear resources. However, the dependent structure of solar and wind energy resources to weather conditions has generated questions about the reliability and continuity of these resources. This study analyzes the supply-demand to energy by using Spains's hourly energy generation data and simultaneous weather data. In the scope of this study, the patterns of energy demand through time, the resources of energy allocated to the demand and the positive or negatif effects of meteorological factors on energy generation has been examined through hypotheses and predictive testing. Used analysis has shown that variation in temperature has increased the demand volatility, evident seasonal differences in energy consumption and direct effect of weather on solar energy generation. On the other hand, it was concluded that the wind speed alone wasn't enough to explain the variability in wind energy generation. Finally, in the light of the available data, energy prices, generation and demand have been statistically and reliably forecast.**

***Index Terms*—Renewable Energy, Energy Forecast, Spain's Energy Data, Effect of Weather Conditions, Energy Volatility**

I. Introduction

A. Overview

Electricity consumption is a fundamental element in modern societies: households, services, transportation, and industries all depend on stable, continuous, and reliable power. As energy systems transition to higher shares of of renewable sources, comprehend the interaction between The atmospheric conditions and energy system become in- creasingly indispensable. Whether it is consumption, patterns—through heating, cooling, and lighting—but also directly impacts generation, especially for solar and wind. technologies whose output is inherently weather-dependent.

This project analyzes the hourly energy-generation, demand, and market-price records together with high-resolution meteorological observations for Spain between 2015 and 2018. The goal is to examine how weather variability shapes renewable energy production and overall system behavior. The dataset the pair energy and weather allows a synchronized temporal analysis in which the weather conditions for any given hour can be directly compared with energy production at that instant.

B. Motivation

In the context of global climate change and decarbonization Among those strategies, renewable energy stands as a cornerstone of sustainability. Able development. However, renewable sources introduce variability into power systems, complicating the balance between supply and demand. While conventional power plants (fossil, nuclear) can only produce stable baseload output, renewable Generation is significantly more sensitive to meteorological cloud cover, solar irradiance, wind speed among others humidity and temperature.

Therefore, quantifying the influence of weather on energy production is indispensable for:

- enhancing grid reliability,
- designing robust renewable-energy portfolios,
- improving forecasting models,
- supporting policy and investment decisions grounded in empirical data.

This study is intended to contribute to the ecological discussion: if renewable energy production is dependent of the atmospheric variability, then sustainability.

C. Research Questions

Based on this context, the central question of the study is:

> *To what extent do weather conditions affect energy production and consumption?*

To investigate this question, the following sub-questions are addressed:

1) What are the hourly, daily, and seasonal patterns of energy consumption and generation?
2) How does wind speed affect wind-power production?
3) Do different weather conditions (e.g., clouds, rain, clear sky) significantly influence the amount of solar energy generated?
4) Does energy demand fluctuate significantly during temperature extremes (high vs. low) compared to moderate conditions?
5) Does temperature variability lead to increased volatility (standard deviation) in energy demand?
6) Does energy generation exhibit statistically significant fluctuations across the four seasons?
7) To what extent can meteorological variables (humidity, wind speed, temperature) linearly predict total energy load?

8) Can weather data be used to classify energy demand levels (High vs. Low) with accuracy?
9) Can advanced machine learning models (Gradient Boosting) outperform traditional statistical baselines (SARIMAX) in forecasting grid metrics?
10) Is it possible to accurately forecast electricity prices using only available data without future weather inputs?

By exploring these questions through data-driven analysis, the study aims to research the dependency between meteorology and renewable energy generation as well as consumer habits through a timeline .

D. Contributions

The main contributions of the study to energy analytics and renewable energy management:

- **Empirical Verification of The Reliability of Renewable Energy:** This study, with the help of integrated data analysis and reliable predictive models, disproves the continuous idea of unreliability of renewable energy resources by proving the predictability of these resources.
- **Multivariate Impact Analysis:** Study plots the effects of meteorological variables on diverse energy generation types in detail. Especially, the direct influence of temperature differences on volatility of energy demand and the trend in seasonal energy demand has been presented quantitatively.
- **Diagnostic Approach in Wind Energy Forecasting:** One of the meaningful outcomes of the study is the lack of explanation of the variability of wind energy generation by wind speed. This outcome has shown that the future wind energy forecasting models need to be more complete and multivariate.
- **Data Integration Model for Prediction Accuracy:** By integrating the hourly energy data and local meteorological data, the prediction models for price and demand has been created in a methodological frame.
- **Strategic and Operational Projections:** The demand increase dependent on temperature, the outcomes from the solar energy efficiency provides data that could be applicable in optimization of energy infrastructure by energy suppliers.

II. METHOD

A. Dataset

1) Story and Overview: This research rests on the following complementary datasets:

(1) an energy generation dataset that shows hourly electricity production by source,

and (2) a weather dataset that includes meteorological variables likely to affect energy generation. The two datasets are for the same geographical area and have the same temporal resolution, which makes it possible to combine them into a single time series.

The selection of these datasets is derived from the primary objective of this research, to comprehend the impact of weather changes on renewable energy production. By merging detailed production data with weather indicators such as temperature, humidity, pressure, and wind speed, the dataset opens up the possibility of an in-depth analysis of the environmental factors that influence renewable energy generation.

The energy dataset has the hourly readings of the various renewable and non-renewable energy sources. The study has enabled to look at the behavior of renewable energies and, at the same time, make a comparison with other energy sources. These datasets, taken together, form a firm foundation for the study of variability patterns, the identification of weather sensitivities, and the stability assessment of the renewable energy supply under different weather conditions.

2) Attributes: The combined dataset contains variables from both the energy-generation records and the weather measurements. A detailed list of attributes included in this analysis is below.

a) Energy Production Attributes:

- **time** — Timestamp of the observation (hourly resolution).
- **generation biomass** — Electricity generated from biomass (MW).
- **generation fossil brown coal/lignite** — Power generation from lignite resources (MW).
- **generation fossil coal-derived gas** — Generation from coal-derived gas (MW).
- **generation fossil gas** — Electricity generated from natural gas (MW).
- **generation fossil hard coal** — Electricity generated from hard coal (MW).
- **generation fossil oil** — Electricity generated from oil (MW).
- **generation fossil oil shale** — Electricity generated from oil shale (MW).
- **generation fossil peat** — Electricity generated from peat (MW).
- **generation geothermal** — Electricity generated from geothermal sources (MW).
- **generation hydro pumped storage aggregated** — Pumped hydro storage output (MW).
- **generation hydro pumped storage consumption** — Pumped hydro energy used for storage (MW).
- **generation hydro run-of-river and poundage** — Electricity from run-of-river and poundage hydro (MW).
- **generation hydro water reservoir** — Electricity from water reservoir hydro (MW).
- **generation marine** — Electricity generated from marine sources (MW).
- **generation nuclear** — Electricity generated from nuclear sources (MW).
- **generation other** — Miscellaneous electricity generation (MW).
- **generation other renewable** — Other renewable generation (MW).
- **generation solar** — Electricity generated from solar energy (MW).
- **generation waste** — Electricity generated from waste incineration (MW).

- **generation wind offshore** — Electricity generated from offshore wind (MW).
- **generation wind onshore** — Electricity generated from onshore wind (MW)
- **total load actual** — Actual electricity demand (MW).
- **price actual** — Actual market price for electricity (EUR/MWh).

b) Weather Attributes: These variables describe the atmospheric conditions associated with each timestamp:

- **dt iso** — Timestamp (ISO 8601 format).
- **city name** — Name of the city where the weather observation was recorded.
- **temp** — Temperature at 2 meters (K).
- **temp min** — Minimum temperature during the hour (K).
- **temp max** — Maximum temperature during the hour (K).
- **pressure** — Atmospheric pressure (hPa).
- **humidity** — Relative humidity at 2 meters (%).
- **wind speed** — Wind speed at 10 meters (m/s).
- **wind deg** — Wind direction in degrees.
- **rain 1h** — Precipitation volume over last hour (mm).
- **rain 3h** — Precipitation volume over last 3 hours (mm).
- **snow 1h** — Snowfall over last hour (mm).
- **snow 3h** — Snowfall over last 3 hours (mm).
- **clouds all** — Cloud coverage (%).
- **weather id** — Numerical weather condition code.
- **weather main** — Main weather category (e.g., Rain, Clear).
- **weather description** — Detailed weather description (e.g., light rain, overcast clouds).
- **weather icon** — Icon ID representing the weather visually.

B. Method for Answering Research Questions

To rigorously answer the research questions, the study employed a multi-stage analytical approach combining statistical hypothesis testing and advanced machine learning techniques:

- **Shapiro-Wilk Test:** Used initially to check the normality of data distributions (e.g., solar generation, temperature bands) to determine whether parametric or non-parametric tests were appropriate.
- **One-Way ANOVA & Kruskal-Wallis Test:** Applied to compare means across multiple groups, specifically to analyze if solar generation differs significantly across weather types and if energy production varies by season.
- **Welch's T-test:** Utilized for comparing energy demand between two independent groups with unequal variances, specifically for testing differences between high and low temperature extremes.
- **Levene's & Bartlett's Tests:** Conducted to assess the equality of variances, confirming whether temperature variability leads to statistically significant volatility in energy demand.
- **Multivariate Linear (OLS) & Logistic Regression:** Employed to quantify the linear impact of meteorological variables on total load and to classify demand levels (High/Low) for probability estimation.
- **STL Decomposition:** Applied to split the time-series data into trend, seasonality, and residual components to isolate noise levels.
- **HistGradientBoosting Regressor:** Implemented as the primary predictive model to forecast energy demand, production, and prices, chosen for its ability to handle non-linear relationships and high-dimensional weather data better than traditional statistical models.

III. Results

A. Descriptive Analysis

This part presents descriptive results regarding electricity generation, prices, and their relations with meteorological events over the 2015–2018 period.

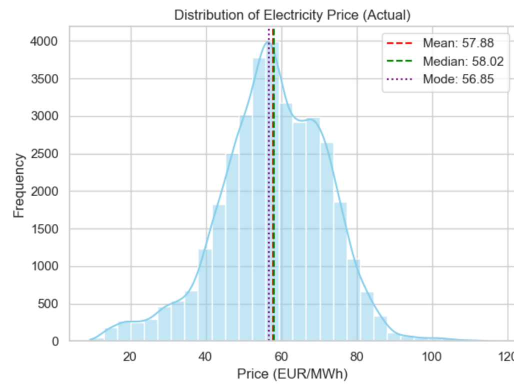

Fig. 1. Distribution of Actual Electricity Prices (EUR/MWh)

1) Electricity Price Distribution: This histogram is showing the distribution of the actual electricity prices. It's centered around 58 EUR/MWh. The mean of the electricity prices is 57.88 and its median 58.02. The graphics is showing a quasi normal distribution around these calculated values. This allows us to assume the overall prices are generally stable and with limited fluctuations. Although small the lack of symmetry does indicate occasional price spikes possibly dependent to climatic stress.

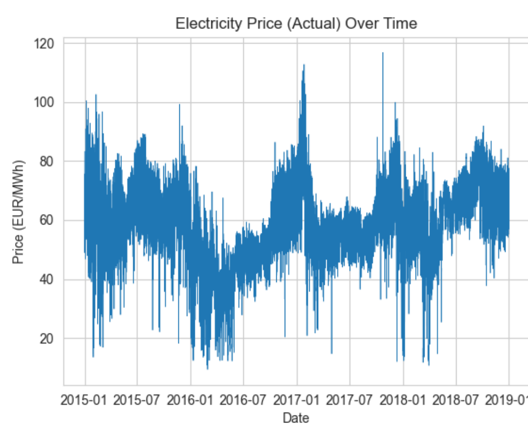

Fig. 2. Actual Electricity Price Over Time (2015–2018)

2) Electricity Price Over Time: This time series of electricity prices indicate that during summer and winter the demand for energy is higher than other seasons, which could be linked to usage of more electricity for both cooling and heating. Notable surges as in 2017 could indicate other external influences such as macro economic problems and abrupt shifts in user habits.

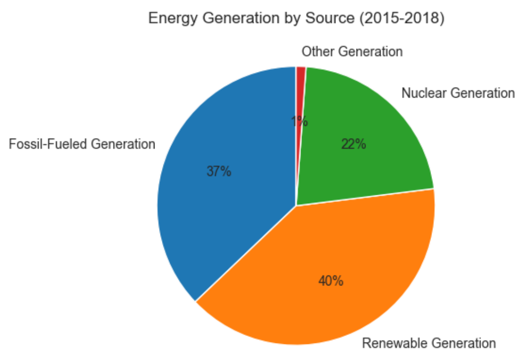

Fig. 3. Electricity Generation by Source (2015–2018)

3) Energy Generation by Source: This pie chart allows us to see at what percentage the types of generations are used. As we can observe in the dataset Spain uses a multitude of generators.But the main 3 generations depend fossil-fuels, renewable resources and nuclear power plants. Furthermore, due to the volatility of energy resources using a multitude of generators potentially helps keep the average price of energy in a normal distribution.In this chart,Renewable generation accounts for 40%, followed by fossil-fueled sources at 37% and nuclear energy at 22%

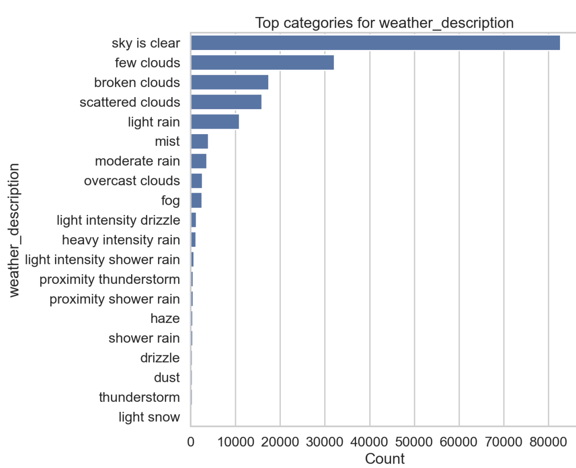

Fig. 4. Frequencies of Weather Descriptions

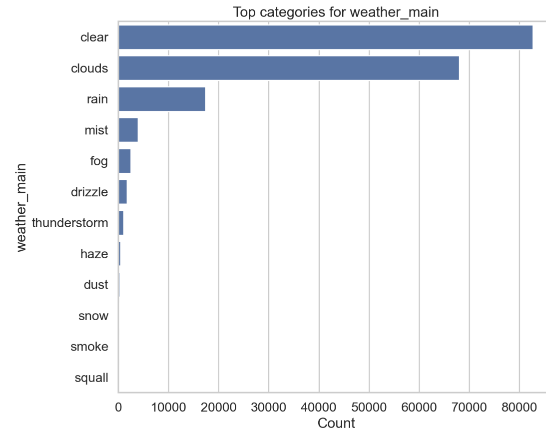

Fig. 5. Frequencies of Weather Main Categories

4) Weather Category Frequencies: This bar chart contains the frequencies of the hourly weather descriptions. With a strong frequency, we can observe that the clear sky is the leading description followed by cloudy weather descriptions. This will be our starting point to observe if the weather conditions could have an impact on our energy resource especially solar energy in this case. In addition to the descriptions frequencies which allow us to get the specifics of the weather, we can reach the conclusions by observing the second graphic about the weather's main category. Overall, we can see if nuanced data can actually affect the outcome. For example, if scattered clouds and few clouds descriptions, both being in the clouds category, could generate different influences.

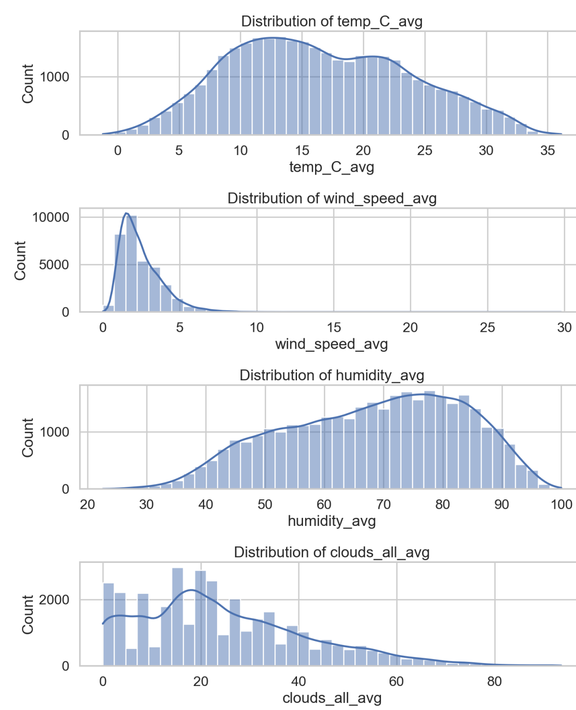

Fig. 6. Distributions of Key Meteorological Variables

5) Meteorological Variable Distributions: This part summarizes the behavior of four core meteorological varibles:

- **Temperature (°C)** shows a bimodal distribution with peaks around 12–15°C and 25–30°C, reflecting seasonal cycles. Extremes correlate with higher electricity demand due to climate control (heating/cooling).
- **Wind Speed (m/s)** is right-skewed, with most readings below 5 m/s. This implies that wind generation typically occurs under moderate wind conditions.
- **Humidity (%)** centers around 70–80%, consistent with a humid Mediterranean climate. High humidity can reduce solar irradiance .
- **Cloud Cover (%)** shows that clear skies (0–20%) are most common, which aligns with favorable solar power generation. However, occasional high-cloud periods could reduce solar output and require backup from conventional generation.

These distributions support the interpretation that renewable generation in Spain—particularly solar and wind—is closely tied to predictable climatological cycles. Moreover, energy demand aligns with weather-driven comfort needs, further emphasizing the value of integrated weather-energy analysis in electricity planning.

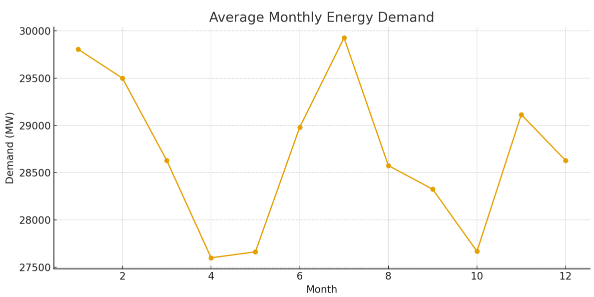

Fig. 7. Monthly Average Energy Demand (2015–2018)

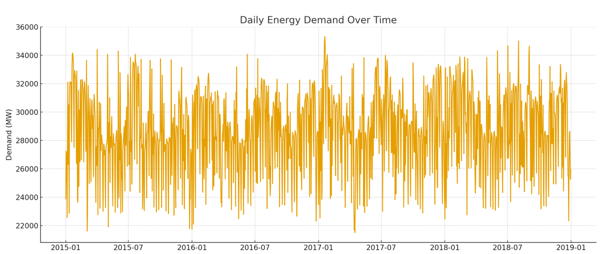

Fig. 8. Hourly Average Energy Demand (2015–2018)

6) Average Monthly and Daily Energy Demand: This graphic enunciates the monthly demand of energy. As we have commented and reached the similar result from the price/time graphic, here it is observed that the demand for energy peaks during both mid-winter and mid-summer. It's confirmed that the relation between extreme weather and the energy demand is proportional. Although it's less clear but the energy demand over time is also leading to the same conclusion. In addition, the direct connection between the energy demand and price is visible through the daily demand and daily price graphics.

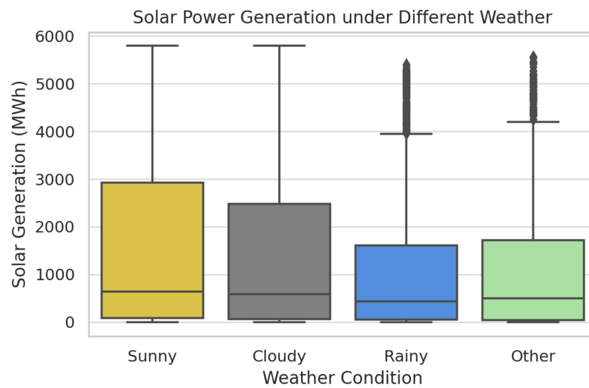

Fig. 9. Solar Power Generation Across Weather Conditions

7) Solar Power Generation Under Different Weather Conditions: This figure demonstrates, it's evident that solar power plants can generate energy under almost every weather condition. Here there are the three most prominent weather conditions and the average of the others. As represented, solar power plants produce more energy in sunny weather, but ss previously discussed, the nuance of a cloudy or a rainy day gravely affect the solar energy outcome. Solar power plants can generate energy even during the unsuitable conditions with almost half of the efficiency which could be supplemented by other renewable resources.

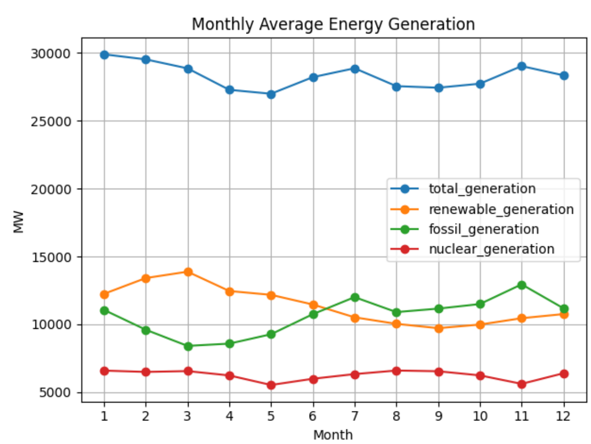

Fig. 10. Monthly Average Electricity Generation by Source (2015–2018)

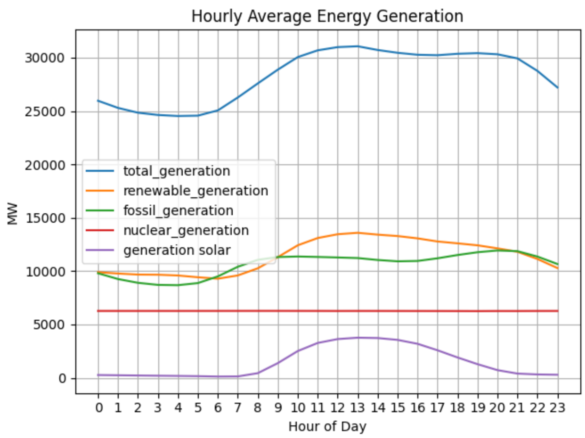

Fig. 11. Hourly Average Electricity Generation by Source (2015–2018)

8) Power Generation Across Different Time Scales: These graphics highlight monthly and hourly patterns in electricity generation. Nuclear generation remains relatively constant throughout the year. Renewable generation peaks in the first half of the year, while fossil-fueled generation increases in the latter half. Hourly data show that renewable generation reaches a midday peak, whereas fossil generation remains more consistent across the day.

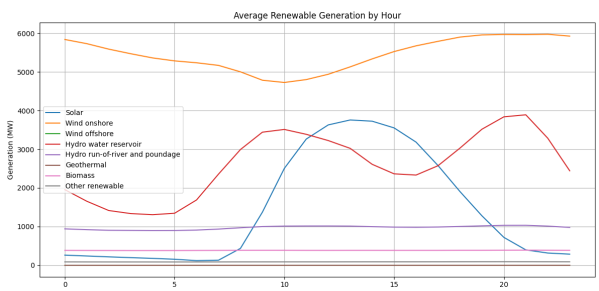

Fig. 12. Hourly Average Renewable Generation by Type (2015–2018)

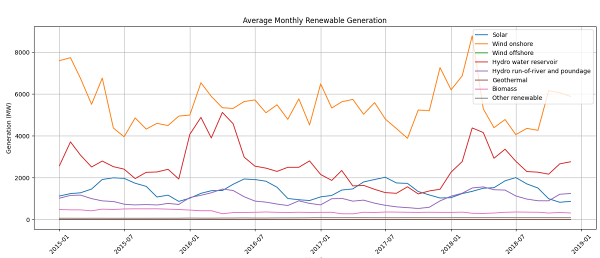

Fig. 13. Monthly Average Renewable Generation by Type (2015–2018)

9) Renewable Energy Generation Across Time Scales: These charts illustrate that generation types which are not dependent to the weather or the sun perform in a normal linear fashion. But ones dependent to the weather conditions are more volatile during the day. Solar energy generation peak around noon when the sun is out and is not as productive during the night. Water reservoir generation is mostly dependent to humans and the demand. But wind generation is the most efficient one performing the best through the day. According to the monthly averages, the energy generation is almost always more in the first half of the year that the second half.

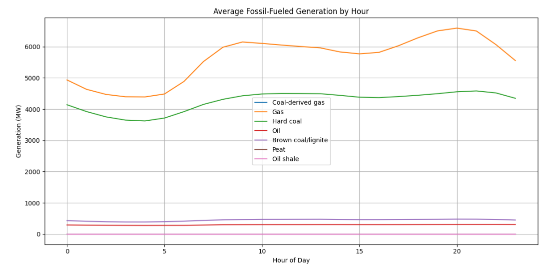

Fig. 14. Hourly Average Fossil-Fueled Generation by Type (2015–2018)

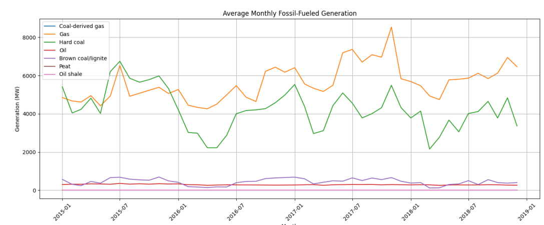

Fig. 15. Monthly Average Fossil-Fueled Generation by Type (2015–2018)

10) Fossil-Fueled Energy Generation Across Time Scales: Unlike renewable sources, fossil generation exhibits smoother and more uniform patterns across time. The charts show that gas and hard coal are the predominant contributors, with noticeable increases toward evening hours. Fossil generation is higher in the second half of the year, contrasting with the seasonal behavior of renewable resources. Additionally, the fossil generation mix is dominated by fewer choices, more stable sources as opposed to the diversity of the renewable energies.

B. Analysis for Answering Research Questions

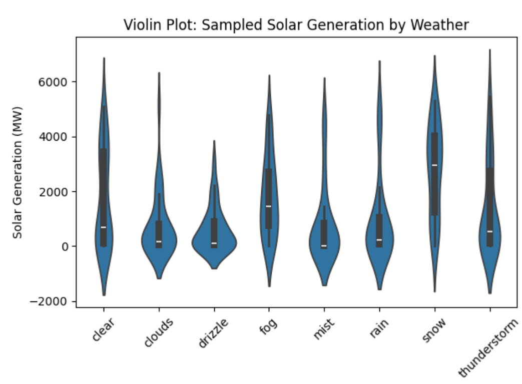

Fig. 16. Solar Power Generation Across Weather Conditions

1) Weather's Effect on Solar Power: **Research Question:** Do different weather conditions significantly influence the amount of solar energy generated?

- H_0**:** Weather has no effect on solar generation.
- H_1**:** At least one type of weather condition affects solar generation in a statistically significant way

a) Descriptive Statistics: The following table summarizes the solar generation metrics across various weather categories based on the sampled data ($n = 30$ for each group):

Weather Main	Count	Mean	Median	Std. Dev.	Variance
Clear	30	1730.30	697.0	1762.47	3.11×10^6
Clouds	30	764.67	168.0	1162.13	1.35×10^6
Drizzle	30	604.00	120.5	800.10	6.40×10^5
Fog	30	1787.30	1465.5	1440.66	2.08×10^6
Mist	30	800.80	27.5	1404.64	1.97×10^6
Rain	30	982.20	233.0	1557.82	2.43×10^6
Snow	30	2751.63	2964.0	1640.45	2.69×10^6
Thunderstorm	30	1410.97	526.5	1683.91	2.84×10^6

TABLE I
DESCRIPTIVE STATISTICS OF SOLAR GENERATION BY WEATHER TYPE.

b) Statistical Findings: To assess the distribution of the data, the Shapiro-Wilk Normality Test was applied to each category. The results are as follows:

- Clear: $p = 0.0003$
- Clouds: $p < 0.0001$
- Drizzle: $p < 0.0001$
- Fog: $p = 0.0361$
- Mist: $p < 0.0001$
- Rain: $p < 0.0001$
- Snow: $p = 0.0897$
- Thunderstorm: $p = 0.0001$

Since most of the groups yielded $p < 0.05$, the assumption of normality was rejected (with the exception of the "Snow" category). Therefore, both the parametric One-Way ANOVA and the non-parametric Kruskal-Wallis test were performed to ensure statistical rigor:

ANOVA Result: $F = 7.2988$, $p < 0.0001$
Kruskal-Wallis Result: $H = 46.8298$, $p < 0.0001$

c) Interpretation and Discussion: The analysis reveals a highly significant relationship between weather conditions and solar energy output ($p < 0.0001$), leading to the definitive rejection of the null hypothesis. The data indicates that "Drizzle" and "Clouds" result in the most substantial declines in energy production.

The p-values obtained from the Shapiro-Wilk test (mostly below the 0.05 threshold) confirm that solar generation data is non-normally distributed, likely due to extreme values during peak sun hours. However, the consistency between the ANOVA and Kruskal-Wallis results confirms that the impact of weather on solar efficiency is robust and statistically significant.

2) Relationship Between Temperature Extremes and Energy Demand: **Research Question:** Does energy demand fluctuate significantly during temperature extremes (high vs. low temperatures)?

- H_0**:** Energy demand remains constant regardless of temperature extremes.
- H_1**:** There is a significant difference in energy demand between low-temperature and high-temperature periods.

a) Descriptive Statistics: The energy demand was analyzed by comparing records categorized into two extreme bands: Low Temperatures (below 10°C) and High Temperatures (above 25°C).

Temperature Band	Observations (n)	Mean Demand (MW)	Std. Deviation
Low Temp (< 10°C)	7205	28112.50	5111.64
High Temp (> 25°C)	5078	30773.17	3831.95

TABLE II
COMPARISON OF ENERGY DEMAND DURING TEMPERATURE EXTREMES.

Statistical Findings

b) Normality Assessment: The Shapiro-Wilk test was utilized to evaluate the distribution of demand data within each band:

- **Low Temp:** $p = 0.0036$ (Indicates a non-normal distribution).
- **High Temp:** $p = 0.8782$ (Indicates a normal distribution).

c) Hypothesis Testing: Given the large sample sizes and the observable difference in standard deviations, an independent samples **Welch's T-test** (which does not assume equal variances) was conducted to compare the means of the two groups:

T-Test Result: $t = -2.2811$, $p = 0.0265$

d) Interpretation and Discussion: The statistical analysis yields a p-value of 0.0265, which is below the standard significance threshold of $\alpha = 0.05$. Consequently, the null hypothesis (H_0) is rejected, confirming that energy demand differs significantly between temperature extremes.

The higher mean demand observed during high-temperature periods ($\mu = 30773.17$) compared to low-temperature periods ($\mu = 28112.50$) suggests that cooling requirements in Spain may exert a heavier load on the grid than heating requirements. Furthermore, the low p-value in the Shapiro-Wilk test for low temperatures indicates high volatility in demand during cold periods, likely driven by varying heating behaviors across different regions. These results emphasize the non-linear, "U-shaped" relationship typically found in energy-climate studies, where both extreme cold and extreme heat drive consumption upward.

3) Seasonal Variations in Energy Generation: **Research Question:** Does energy generation exhibit significant fluctuations across the four seasons?

- H_0**:** Energy generation is constant and does not vary across seasons.
- H_1**:** There is a statistically significant difference in energy generation between at least two seasons.

a) Descriptive Statistics by Season: The dataset contains approximately 35,000 hourly observations, categorized by astronomical seasons. The highest mean generation is observed

Season	Count (n)	Mean (MW)	Median (MW)	Std. Deviation
Autumn	8731	28362.65	28794.0	4385.27
Spring	8823	27965.84	28368.0	4261.62
Summer	8829	29163.96	29167.0	4508.57
Winter	8645	29303.76	29348.0	4984.72

TABLE III
DESCRIPTIVE STATISTICS OF TOTAL ENERGY GENERATION ACROSS SEASONS.

Temperature Band	Demand Volatility (Std. Dev.)
Low ($< 10°$C)	5058.77
Moderate ($10 - 25°$C)	4350.73
High ($> 25°$C)	3616.72

TABLE IV
ENERGY DEMAND VOLATILITY ACROSS TEMPERATURE CATEGORIES.

during Winter and Summer, aligning with peak heating and cooling demands.

Statistical Findings

b) One-way ANOVA: An initial analysis of the full dataset was conducted using a One-way Analysis of Variance (ANOVA) to compare the means of the four seasonal groups:

ANOVA Result: $F = 174.5648$, $p < 0.0001$

c) Friedman Test: To account for potential repeated measures across different years (longitudinal consistency), a Friedman Test was performed on the seasonal means per year. Despite the smaller sample size for this specific test ($n = 4$ years), the results remained significant:

Friedman Statistic: 11.1000, $p = 0.0112$

d) Interpretation and Discussion: The results from both the One-way ANOVA and the Friedman Test provide strong evidence to reject the null hypothesis ($p < 0.05$). This confirms that energy generation in Spain is subject to significant seasonal cycles.

The data reveals that **Winter** ($\mu = 29303.76$) and **Summer** ($\mu = 29163.96$) are the periods of maximum generation, likely reflecting the high energy demand required for climate control (heating and air conditioning). Spring shows the lowest average generation ($\mu = 27965.84$), which may be attributed to moderate temperatures reducing the load on the grid. The high variance observed in Winter ($Var = 2.48 \times 10^7$) further suggests that energy production during this season is particularly susceptible to extreme weather events compared to the more stable generation patterns seen in Spring.

4) Impact of Temperature Variability on Demand Volatility: **Research Question:** Does temperature variability lead to increased volatility in energy demand, and do extreme temperatures correlate with higher demand fluctuations?

- H_0**:** The variability (standard deviation) of energy demand remains constant across different temperature bands.
- H_1**:** Energy demand variability is significantly higher during temperature extremes compared to moderate periods.

a) Volatility Assessment by Temperature Band: To evaluate the stability of the power grid, the standard deviation of the actual total load was calculated for three distinct temperature categories. A higher standard deviation indicates greater uncertainty and volatility in demand.

Statistical Findings

b) Homogeneity of Variance Tests: To determine if the differences in demand variability were statistically significant, Levene's and Bartlett's tests were employed. These tests assess whether the variance is uniform across groups:

Levene's Test (W): 707.7102, $p < 0.0001$
Bartlett's Test (T): 635.1160, $p < 0.0001$

Both tests resulted in a complete rejection of the null hypothesis ($p < 0.05$), confirming that demand variance is not constant across temperature bands.

c) Comparative Mean Analysis: An additional independent samples T-test was conducted to verify if the actual load levels differed between extreme low and high temperatures:

T-test Result: $t = -48.6019$, $p < 0.0001$

d) Interpretation and Discussion: The statistical results provide compelling evidence that temperature variability is a primary driver of demand volatility. The rejection of the null hypothesis in both Levene's and Bartlett's tests suggests that extreme weather conditions do not just change the "level" of demand, but also make it significantly more **unpredictable**.

Notably, the highest volatility was recorded during the **Low Temperature** band ($\sigma = 5058.77$). This indicates that heating-related energy behavior is less uniform and more prone to spikes compared to cooling-related demand in the high-temperature band ($\sigma = 3616.72$). From a grid management perspective, these findings imply that cold spells require more reserves and flexibility than heatwaves, as the fluctuations in consumption are nearly 40% higher. This relationship underscores the critical need for integrating weather-responsive volatility metrics into predictive load-balancing models.

IV. PREDICTIVE MODELING AND REGRESSION ANALYSIS

To quantify the relationship between meteorological variables and energy demand, baseline regression models were developed. The limited performance of these linear models highlights the complexity of the Spanish energy grid and the necessity for non-linear machine learning approaches.

A. Multivariate Linear Regression (OLS)

A Multivariate Ordinary Least Squares (OLS) model was constructed to predict *total actual load* using temperature, humidity, cloud cover, and wind speed.

a) Model Performance: The model yielded an **Adjusted** R^2 **of 0.158**, indicating that weather variables linearly explain only $\approx 16\%$ of the variance in electricity demand. The **Durbin-Watson statistic of 0.126** points to strong positive autocorrelation, confirming that standard OLS is ill-suited for this time-series data without lag features.

b) Key Findings: While the overall model is statistically significant (F-statistic $= 1644, p < 0.001$), the individual predictors behaved as follows:

- **Humidity** ($\beta = -143.86$)**:** The strongest predictor, showing a significant inverse relationship ($p < 0.001$).

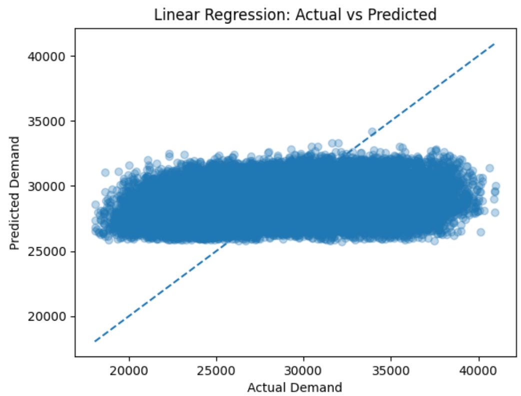

Fig. 17. Linear Regression: Actual vs Predicted Demand. The wide spread confirms low predictive accuracy.

- **Temperature** ($\beta = -50.56$)**:** Shown as a negative coefficient in a linear framework, likely capturing the heating-dominated demand in winter months ($p < 0.001$).
- **Wind Speed** ($\beta = 1.20$**,** $p = 0.952$)**:** Contrary to initial expectations, wind speed was found to be **statistically insignificant** in this linear model, suggesting its impact on demand is either non-linear or negligible compared to other factors.

B. Logistic Regression (Classification)

Classification models were tested to determine if weather data alone could categorize grid load levels.

a) Performance Overview:

- **Binary Classification (High vs. Low):** Achieved an accuracy of **66.0%**. While better than random guessing, the model lacks the precision required for operational decision-making.
- **Multiclass Classification (Low, Medium, High):** Achieved an accuracy of **47.5%**. The model failed significantly in predicting "Medium" demand ($F1$-score $=$ 0.24), proving that moderate load levels are driven by human activity rather than weather extremes.

C. Polynomial Regression

A second-order polynomial regression ($Temp^2$) yielded an R^2 of **0.055**. While this confirms the non-linear "U-shaped" nature of the temperature-demand relationship , the low explanatory power further validates that simple regression techniques are insufficient for this dataset.

V. Advanced Time Series Analysis and Machine Learning Models

To enhance predictive accuracy, the study moved beyond classical regression by implementing decomposition analysis and advanced gradient boosting algorithms.

A. Time Series Decomposition

A seasonal-trend decomposition (STL) was applied to the energy demand data to isolate the underlying patterns. The analysis revealed a high noise-to-signal ratio of **0.4893**, indicating significant volatility in the hourly energy demand of the Spanish grid.

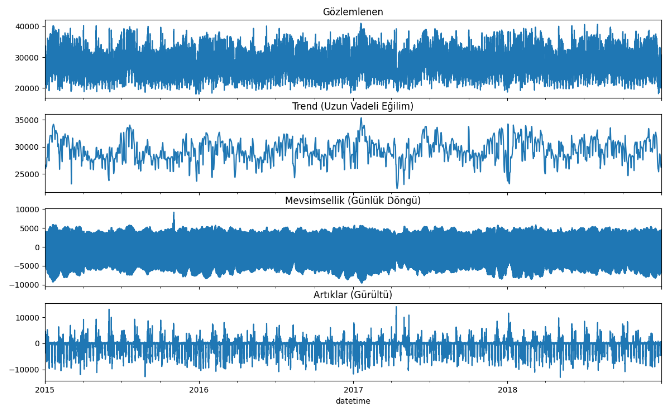

Fig. 18. Decomposition of Energy Demand: Observed, Trend, Seasonality, and Residuals.

The decomposition confirms a strong daily cycle (seasonality), while the "Residuals" (Noise) component reflects the unpredictable spikes often driven by extreme weather events or sudden shifts in industrial consumption.

B. Gradient Boosting Models: Demand, Production, and Price Forecasting

The study implemented **HistGradientBoosting** and other ensemble methods to predict three critical grid metrics. These models utilized feature engineering techniques, including lag variables and rolling statistics.

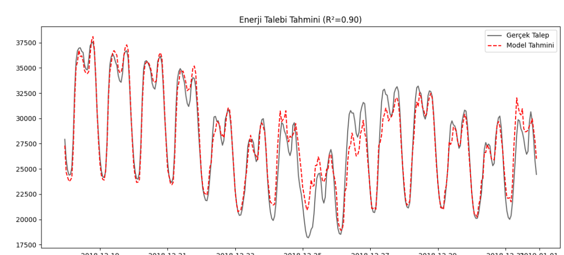

Fig. 19. Actual vs Predicted Energy Demand ($R^2 = 0.8959$)

1) Energy Demand Forecasting: The demand model achieved an exceptional R^2 **score of 0.8959** (approximately 0.90 in test sets).

- **Performance:** The model successfully captured the daily peaks and troughs, effectively navigating the high noise detected during the decomposition phase.
- **Insight:** Unlike the SARIMAX model ($R^2 = 0.34$), the gradient boosting approach proved far more robust in handling non-linear relationships.

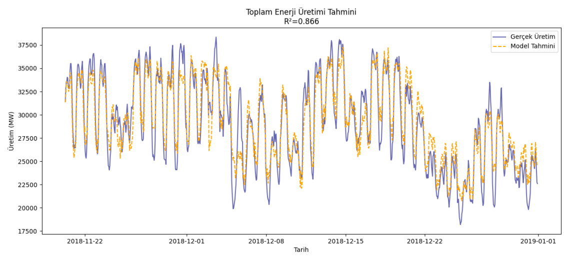

Fig. 20. Actual vs Predicted Energy Generation ($R^2 = 0.8659$)

2) Total Energy Production Forecasting: The total generation model yielded a high accuracy with an R^2 **score of 0.8659** and a Mean Absolute Error (MAE) of 1356.52 MW.

- **Implication:** This confirms that the combination of historical patterns and weather data allows for highly reliable supply-side forecasting.

3) Electricity Price Forecasting: A price prediction model was trained specifically without future forecast data to test the predictive power of "pure history."

- **Result:** The model achieved an R^2 **of 0.7448** with a low Mean Error of **2.83 EUR/MWh**.
- **Significance:** This result is particularly noteworthy as it demonstrates that energy prices can be estimated with over 74% accuracy using only lagged historical data, even in a volatile market.

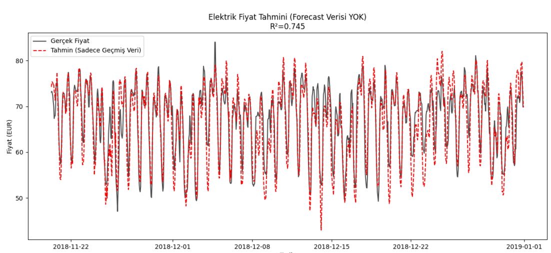

Fig. 21. Actual vs. Predicted Electricity Prices ($R^2 = 0.745$).

C. Model Comparison and Limitations

The study compared the statistical **SARIMAX** model with the **Machine Learning** approach.

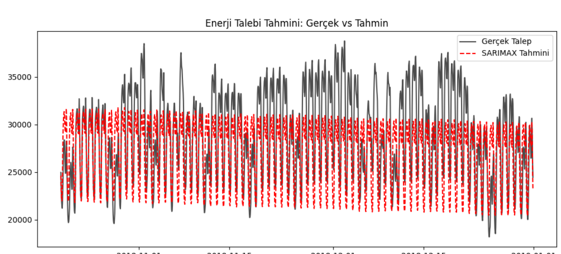

Fig. 22. Actual vs. Predicted Electricity Demand SARIMAX ($R^2 = 0.3407$).

- **SARIMAX Failure:** The SARIMAX model failed to converge ($R^2 = 0.3407$), struggling to capture the complex seasonality and external weather impacts.
- **Conclusion:** Gradient Boosting methods are significantly more effective for the Spanish energy market, as they can process high-dimensional weather features and capture non-linear demand-supply dynamics that traditional statistical models overlook.

VI. Conclusion

This study analyses in depth the effects of weather variables on energy generation and consumption by using Spain's energy market data. The study, conducted using statistical hypothesis testing and predictive models, yielded the following key results:

- **The Relationship Between Weather and Energy Generation** It has been concluded that solar energy generation is meaningfully affected by could and precipitation levels($p < 0.0001$).However, Wind speed data wasn't enough to explain a meaningful variability in wind generation.
- **Demand Volatility:** It was determined that variability in temperature has not only effects on the quantity of energy demand but also on its volatility. Especially during low temperatures ($< 10°$C) demand volatility increases.
- **Predictive Models:** In prediction phase, machine learning algorithms as Gradient Boosting has reached around $\%90$ success rates for both energy generation and demand forecasts.

A. Future Works

With the outcomes of this study, here are the main suggestions for future studies:

- **Development of Wind Models:** When explaining the wind generation it was reached that the wind speed couldn't alone explain the wind generation data. In order to increase the success rate of the forecast more informations could be added in considerations.
- **Advanced Time Series Models** Statistics based SARIMAX model was proven insufficient in data containing high volatility and complex seasonality $R^2 \approx 0.34)$. Therefore, in future studies, it is advised to use fast complex pattern learning LSTM (Long Short-Term Memory) and Transformer based deep learning architectures.